

CS2301 Assignment - XML Validation

Application Domain: E-commerce Product Catalog

Name: _____

Roll No: _____

Date: _____

XML Code (products.xml)

```
<?xml version="1.0" encoding="UTF-8"?>
<products xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="products.xsd">
  <product id="P001">
    <name>Wireless Mouse</name>
    <category>Electronics</category>
    <price>25.99</price>
    <stock>150</stock>
  </product>
  <product id="P002">
    <name>Mechanical Keyboard</name>
    <category>Electronics</category>
    <price>89.50</price>
    <stock>75</stock>
  </product>
  <product id="P003">
    <name>Office Chair</name>
    <category>Furniture</category>
    <price>199.99</price>
    <stock>20</stock>
  </product>
  <product id="P004">
    <name>Standing Desk</name>
    <category>Furniture</category>
    <price>350.00</price>
    <stock>15</stock>
  </product>
  <product id="P005">
    <name>Noise Cancelling Headphones</name>
    <category>Audio</category>
    <price>299.99</price>
    <stock>40</stock>
  </product>
  <product id="P006">
    <name>Bluetooth Speaker</name>
    <category>Audio</category>
    <price>45.00</price>
    <stock>100</stock>
  </product>
  <product id="P007">
    <name>USB-C Hub</name>
    <category>Accessories</category>
    <price>35.50</price>
    <stock>200</stock>
  </product>
  <product id="P008">
    <name>Webcam 1080p</name>
    <category>Electronics</category>
    <price>59.99</price>
    <stock>60</stock>
  </product>
```

```

<product id="P009">
  <name>Laptop Stand</name>
  <category>Accessories</category>
  <price>29.99</price>
  <stock>120</stock>
</product>
<product id="P010">
  <name>Desk Lamp</name>
  <category>Home Office</category>
  <price>39.99</price>
  <stock>85</stock>
</product>
</products>

```

XSD Code (products.xsd)

```

<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

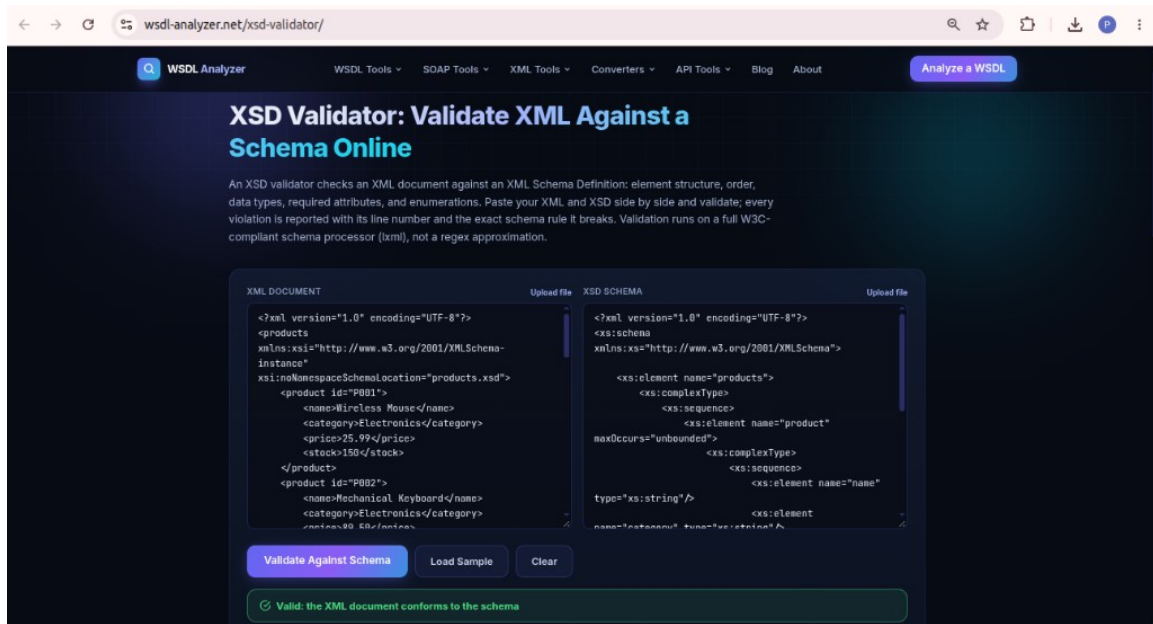
  <xs:element name="products">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="product" maxOccurs="unbounded">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="name" type="xs:string"/>
              <xs:element name="category" type="xs:string"/>
              <xs:element name="price" type="xs:decimal"/>
              <xs:element name="stock" type="xs:integer"/>
            </xs:sequence>
            <xs:attribute name="id" type="xs:ID" use="required"/>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>

</xs:schema>

```

Screenshots

1. Successful Validation



2. Error Validation

